RANSAC

1. Algorithm
    1. Sample (randomly) the number of points s required to fit the model.
    2. Solve for model parameters using samples.
    3. Score by the fraction of inliers within a preset threshold of the model
    4. Repeat 1-3 until the best model is found with high confidence
2. Choosing parameters
    1. Number of algorithm iterations N
        1. choose N so that, with probability p (e.g. $p = 0.99$), at least one random sample is free from outliers (outlier ratio: e)
    2. Number of sampled points s.
        1. Minimum number needed to fit the model.
    3. Distance threshold δ
        1. Choose δ so that a good point with noise is likely (e.g. prob=0.99) within threshold.
    4. How many iterations do I need? $N = \log(1-p)/\log(1-(1-\exp)^s)$
3. Applying RANSAC for feature matching between two images [4]:
    1. recall the RANSAC steps are:
        1. Select four feature pairs (at random), p_i from image 1, p_i^1 from image 2.
        2. Compute homography H (exact). Use the function `est_homography`.
        3. Compute inliers where $SSD(p_i^1, Hp_i) < \text{thresh}$. Here, Hp_i computed using the `apply_homography` function given to you. SSD: sum of squares difference.
        4. Repeat the last three steps until you have exhausted N_{max} number of iterations (specified by user) or you found more than percentage of inliers (say 90% for example).
        5. Keep largest set of inliers.
        6. Re-compute least-squares $\hat{H}$ estimate on all of the inliers. Use the function `est_homography` given to you.
4. Applications of RANSAC [2]
    1. Multiple areas in computer vision require robust estimation techniques
        1. Camera pose estimation
        2. Triangulation
        3. 2D registration
        4. 3D registration (also computer graphics)
        5. 3D model fitting (computer graphics)
        6. Line detection
        7. Rotation averaging
        8. Bundle adjustment
    2. Techniques:
        1. RANSAC
        2. Iteratively reweighted least squares (robust loss functions)
    3. Camera motion estimation
        1. Needs to be robust against mismatches.
        2. All feature matches need to follow the same motion.
    4. Steps overview:
        1. Random sampling of 2 points
        2. Calculate line model from this 2 data points.
        3. Calculate residual error for each data point (normal distance to line).
        4. Select data points that support current hypothesis.
        5. Repeat sampling.
        6. Until a hypothesis with a maximum number of inliers has been found.
    5. RANSAC algorithm:
        1. Initial: Let A be a set of N data points.
        2. Repeat:
            1. Randomly select a sample of s data points from A.
            2. Fit a model to these points.

        3. Compute the distance of all other points to this model.
        4. Construct the inlier set (i.e. count the number of data points whose distance from the model < threshold d)
        5. Store these inliers.
    3. Until maximum number of iterations is reached.
    4. The model with maximum number of inliers is chosen as the solution to the problem.
    5. Re-estimate the model using all the inliers.
6. How many iterations of RANSAC?
    1. The number of iterations N which is necessary to guarantee that at least a single correct solution is found can be computed by:
    2. $N = \frac{\log(1-p)}{\log(1-(1-\epsilon)^s)}$
    3. s is the number of data points from which a model can be minimally computed.
    4. ϵ is the percentage of outliers in the data (can only be guessed)
    5. p is the requested probability of success.
    6. Example: p = 0.99, s = 5, $\epsilon = 0.5 \rightarrow N = 145$
    7. RANSAC is an iterative method and it is non deterministic. It returns different solutions on different runs.
    8. For reasons of reliability, in many practical applications N is usually multiplied by a factor of 10.
    9. More advanced implementations of RANSAC estimate the fraction of inliers adaptively, for every iterations, and use it to update N.
    10. N is exponential in the number of data points s necessary to estimate the model.
    11. Therefore, it is very important to use a minimal parametrization of the model.

number of point (s) p=0.99, e=0.5	**8**	**7**	**6**	**5**	**4**	**2**	**1**
Number of iterations (N)	1177	587	292	145	71	16	7

2. Implementation [1]
    1. Overview
        1. RANSAC stands for Random Sample Consensus, it is a popular algorithm to robustly fit a model (like a line or plane) when the data contains a large number of outliers.
        2. Key steps:
            1. Randomly select a small subset of data to form a model.
            2. Determine how many points from the entire dataset fit the model (inliers).
            3. Update the best model if the new one has more inliers.
            4. Adaptively update the number of iterations needed based on the ratio of inliers found.
    2. Synthetic Data Generation
        1. This function generates a set of N 2D points, consisting of:
            1. Inlier points that lie roughly along a line passing through a reference point p in the direction of d.
            2. Outlier points scattered randomly within a bounding rectangle $[0, Width] \times [0, Height]$.
        2. Steps:
            1. Normalize direction vector $\vec{d}$ as:
                1. $\vec{d} = \frac{d}{\|\vec{d}\|}$
            2. Partition N into inliers and outliers
                1. $N_{inlier} = floor(N \times \text{inlier\_ratio})$
                2. $N_{outlier} = N - N_{inlier}$
            3. Generate inliers along a line
                1. Define p as base point.
                2. select $t \sim Uniform(-\frac{diag}{2}, \frac{diag}{2})$, where:
                3. $diag = \sqrt{Width^2 + Height^2}$.
                4. Compute each inlier point: $\vec{q} = \vec{p} + t\vec{d}$
                5. Add Gaussian noise $\mathcal{N}(0, 1)$ to both coordinates. $q[0] := q[0] + \mathcal{N}(0, 10)$, $q[1] := q[1] + \mathcal{N}(0, 10)$
            4. Generate outliers
                1. For each of the $N_{outlier}$ points, sample randomly:
                2. $x \sim Uniform(0, Width)$
                3. $y \sim Uniform(0, Height)$.

5. Return Combined array of shape (N, d), d=2.

3. Mathematical explanation of iteration_number(confidence, sample_size, inliers, N)
    1. This function computes the number of RANSAC iterations required to ensure, with a given confidence level, that at least one randomly selected subset consists entirely of inliers.
    2. Steps:
        1. Definitions:
            1. Inlier ratio (w): The fraction of points that are inliers.
            2. Sample size (s): The minimum number of points needed to estimate the model.
            3. Confidence (p): The probability that at least one sample contains only inliers.
            4. The probability of picking an inlier at random is:
            5. $w = \frac{inliers}{N}$,
            6. If (inliers=0), then (w=0), meaning RANSAC cannot reliably find a correct model. In this case, the function returns $(\inf)$ (indicating an unbounded number of iterations might be required).
        2. Probability of a good sample
            1. The probability of a randomly chosen sample of size (s) consists only of inliers is w^s.
        3. Probability of All bad samples
            1. If we run (k) iterations, the probability that non of these (k) samples contain only inliers is: $(1 - w^s)^k$.
        4. Confidence constraint
            1. To ensure a confidence level (p), we require $(1 - w^s)^k \leq 1 - p$.
        5. Solving for (k)
            1. Taking the natural logarithm of both sides:
            2. $k \ln(1 - w^s) \leq \ln(1 - p)$.
            3. Thus $k \geq \frac{\ln(1-p)}{\ln(1-w^s)}$.
            4. Since RANSAC requires an integer number of iterations, we take the ceiling: $k = ceil(\frac{\ln(1-p)}{\ln(1-w^s)})$.
        6. Return value
            1. If $inliers == 0$, we return (inf) since no inlier-based model can be drawn.
            2. If the denominator $(ln(\ldots))$ is zero or extremely small, we also return (inf).
            3. Otherwise, we return the ceiling of the above fraction, indicating the required iterations to reach the given confidence level.

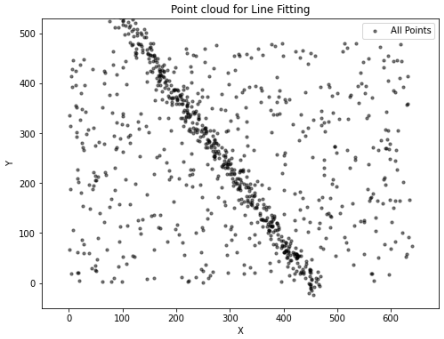
4. Code:

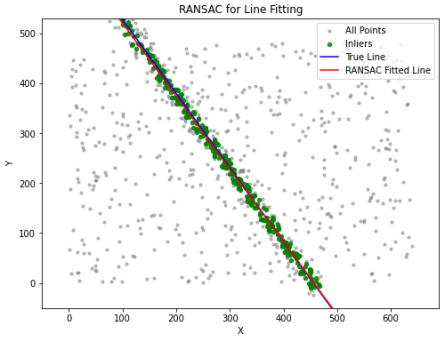
    1. **Initialize the random seed** for reproducibility.
    2. **Generate synthetic data** of size 1000, with 50% inliers.
    3. **Apply RANSAC** to fit a line.
    4. **Refine the best line** using all inliers.

```python
# -*- coding: utf-8 -*-
"""
Created on Sun Mar  8 18:22:50 2026

@author: reina
"""

import numpy as np
import matplotlib.pyplot as plt
import random
import math


def init_random(seed=None):
    if seed is not None:
        np.random.seed(seed)
        random.seed(seed)


def generate_data(N, p, direction, inlier_ratio=0.5):
    WIDTH, HEIGHT = 640, 480
    p = np.array(p, dtype=float)
    direction = np.array(direction, dtype=float)
    direction /= np.linalg.norm(direction)

    N_inlier = int(N * inlier_ratio)
    N_outlier = N - N_inlier

    points = []

    diag = np.sqrt(WIDTH**2 + HEIGHT**2)
```

```
    for _ in range(N_inlier):
        t = (random.random() * diag) - diag / 2.0
        q = p + t * direction
        q[0] += np.random.normal(0.0, 10.0)
        q[1] += np.random.normal(0.0, 10.0)
        points.append(q)

    for _ in range(N_outlier):
        points.append((random.random() * WIDTH, random.random() * HEIGHT))

    return np.array(points, dtype=float)


def iteration_number(confidence, sample_size, inliers, N):
    if inliers == 0:
        return float('inf')
    ratio = inliers / float(N)
    top = math.log(1.0 - confidence)
    bottom = math.log(1.0 - ratio**sample_size)
    return math.ceil(top / bottom) if bottom != 0 else float('inf')


def fit_line_two_points(p1, p2):
    x1, y1 = p1
    x2, y2 = p2
    dx, dy = x2 - x1, y2 - y1
    norm = math.sqrt(dx**2 + dy**2)
    if norm < 1e-8:
        return None
    a, b = -dy / norm, dx / norm
    c = -a * x1 - b * y1
    return a, b, c


def distance_point_line(line_params, point):
    a, b, c = line_params
    x, y = point
    return abs(a*x + b*y + c)


def ransac_line(points, max_iterations=2000, sigma=10.0, confidence=0.95):
    best_inliers_count = 0
    best_line, best_inliers_list = None, []

    for _ in range(max_iterations):
        idx1, idx2 = np.random.choice(len(points), 2, replace=False)
        line_params = fit_line_two_points(points[idx1], points[idx2])
        if line_params is None:
            continue

        inliers = [pt for pt in points if distance_point_line(line_params, pt) < sigma]

        if len(inliers) > best_inliers_count:
            best_inliers_count, best_line, best_inliers_list = len(inliers), line_params, inliers
            max_iterations = min(max_iterations, iteration_number(confidence, 2, best_inliers_count,
len(points)))
            print(max_iterations)


    return best_line, np.array(best_inliers_list)


if __name__ == '__main__':
    init_random(seed=0)

    N = 1000
    p = (240.0, 320.0)
    alpha = random.random() * 2.0 * np.pi
    dir_ = (np.cos(alpha), np.sin(alpha))
    points = generate_data(N, p, dir_, inlier_ratio=0.5)

    fig, ax = plt.subplots(figsize=(8,6))
    ax.scatter(points[:,0], points[:,1], s=10, c='k', alpha=0.5, label='All Points')


    ax.set_xlim([-50, 690])
    ax.set_ylim([-50, 530])
    ax.set_xlabel('X')
    ax.set_ylabel('Y')
    ax.set_title('Point cloud for Line Fitting')
    ax.legend()
    plt.show()

    best_line, inliers = ransac_line(points)
    print(f"Best line: a={best_line[0]:.3f}, b={best_line[1]:.3f}, c={best_line[2]:.3f}")
    print(f"Number of inliers: {len(inliers)} out of {len(points)}")

    fig, ax = plt.subplots(figsize=(8,6))
    ax.scatter(points[:,0], points[:,1], s=10, c='gray', alpha=0.5, label='All Points')
    ax.scatter(inliers[:,0], inliers[:,1], s=20, c='green', alpha=0.8, label='Inliers')
```

```
p1 = (p[0] - 1000*dir_[0], p[1] - 1000*dir_[1])
p2 = (p[0] + 1000*dir_[0], p[1] + 1000*dir_[1])
ax.plot([p1[0], p2[0]], [p1[1], p2[1]], c='blue', label='True Line')

a, b, c = best_line
x_vals = np.linspace(0, 640, 2)
y_vals = (-a*x_vals - c) / b
ax.plot(x_vals, y_vals, c='red', label='RANSAC Fitted Line')

ax.set_xlim([-50, 690])
ax.set_ylim([-50, 530])
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_title('RANSAC for Line Fitting')
ax.legend()
plt.show()
```





3. Generalization [3]
    1. Expectation Maximization (EM) can also be used, however EM is sensitive to initial condition, not robust to outliers if initial condition is far from the true one.
    2. Solutions:
        1. GNC algorithm
        2. RANSAC algorithm
    3. EM is a simple method for model fitting in the presence of outliers (very noisy points or wrong data).
    4. It can be applied to all sorts of problems where the goal is to estimate the parameters of a model from the data (e.g. camera calibration, Structure from Motion, DLT, PnP, P3P, Homography, etc.)
    5. RANSAC:
        1. How many iterations does RANSAC need?
        2.

References:

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4. https://cmsc426.github.io/pano/